Sequence Modeling of Wrist-Worn Actigraphy for Differentiating Bipolar and Unipolar Depressive Episodes

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Abstract. Bipolar Disorder is routinely mistaken for Major Depressive Disorder, with patients waiting nearly a decade before receiving a correct diagnosis. Both conditions present identically during depressive episodes, yet antidepressants prescribed to unrecognized bipolar patients can precipitate dangerous manic switching. We investigate whether wrist-worn actigraphy can passively distinguish these conditions using the Depresjon dataset (55 participants: 32 healthy controls, 8 bipolar, 15 unipolar). A hybrid 1D-CNN-LSTM architecture operating directly on raw minute-level activity counts is evaluated through four systematic approaches: baseline deep learning with SMOTE, balanced downsampling, four alternative neural architectures, and a diagnostic phase combining classical machine learning with statistical hypothesis testing. Motor activity patterns do not meaningfully differ between bipolar and unipolar depression during depressive episodes. The best reliable accuracy is 63.4% with 48-hour windows, below the 65.2% majority-class baseline, and statistical testing confirms negligible group differences (Mann-Whitney $p = 0.776$, Cohen's $d = -0.060$). A 5-feature logistic regression matches a 223K-parameter deep network at 60.87% LOOCV accuracy. These findings constitute a rigorous negative result, ruling out wrist activity variability as a standalone diagnostic biomarker and redirecting future work toward multi-modal and longitudinal approaches.

Keywords: bipolar disorder, unipolar depression, actigraphy, CNN-LSTM, sequence modeling, wearable sensors, differential diagnosis, negative result

1 Introduction

Bipolar Disorder (BD) affects approximately 1 to 2% of the global population [1], yet epidemiological studies consistently find that patients wait nearly a decade before receiving a correct diagnosis [2]. The core reason is structural. Depressed patients seek clinical help, while hypomanic and manic episodes frequently go unreported or are experienced as periods of elevated productivity. A clinician encountering a patient in the middle of a depressive episode therefore has no direct behavioral evidence to separate BD from ordinary Major Depressive

Disorder (MDD). This diagnostic confusion carries serious clinical consequences: antidepressant monotherapy prescribed to an unrecognized bipolar patient can trigger manic switching, rapid cycling, and mixed states, all of which worsen long-term prognosis [3].

Current diagnostic practice relies on structured clinical interviews, self-report questionnaires such as the PHQ-9, MADRS, and HCL-32, and careful longitudinal observation. All of these approaches share a common vulnerability: they depend on the patient's retrospective account of their own mood states. Recall bias is well-documented in this population, and the cognitive impairments associated with depressive episodes themselves further compromise accurate self-reporting. There is therefore a compelling clinical motivation for objective, passive monitoring tools that operate independently of patient recall.

Wrist-worn actigraphy offers one such tool. Modern consumer-grade wearables continuously record movement-derived activity counts at minute-level resolution, generating multi-day behavioral time series without any active participation from the wearer. The psychiatric literature has long established that both bipolar and unipolar depression are accompanied by characteristic disruptions to rest-activity cycles and circadian rhythms [4, 5]. If these disruptions carry distinct temporal signatures across the two conditions, an automated classifier operating on raw actigraphy data could in principle assist clinicians in making the differential diagnosis earlier and more reliably.

Prior computational work on this problem has largely operated within the classical machine learning paradigm, applying support vector machines, random forests, and related methods to manually extracted statistical summaries of actigraphy signals [6, 7]. These features capture average activity levels reasonably well but miss subtler temporal structure: fragmented sleep bouts, irregular rest-activity transitions across days, and multi-day rhythmic drift. We argue that a deep sequence model that processes the raw time series directly, without prior feature engineering, is better positioned to discover the kinds of multi-scale temporal patterns that distinguish the two conditions [8].

In this paper, we ask a concrete empirical question: can a deep sequence model learn to distinguish bipolar from unipolar depression directly from raw wrist actigraphy, without manual feature engineering. We propose a hybrid 1D-CNN-LSTM architecture that pairs convolutional layers for localized kinetic pattern extraction with a recurrent LSTM module for multi-day information integration. We evaluate this architecture on the publicly available Depresjon dataset [6] through a systematic four-approach investigation that progressively probes the signal from multiple angles: deep learning with synthetic oversampling, balanced training via downsampling, alternative neural architectures, and a diagnostic phase combining classical machine learning with direct statistical hypothesis testing.

The remainder of this paper is structured as follows. Section 2 surveys related work across three domains: motor activity as a psychiatric biomarker, deep learning for wearable time-series, and the psychiatric literature on circadian disruption in mood disorders. Section 3 describes the dataset, preprocessing

pipeline, and base model architecture. Section 4 presents all four experimental approaches with full results and analysis. Section 5 draws conclusions, addresses open questions, and discusses the ethical implications of deploying automated diagnostic tools in mental health contexts.

2 Related Work

2.1 Motor Activity as a Psychiatric Biomarker

The use of actigraphy to characterize psychiatric conditions has a well-established empirical foundation. Garcia-Ceja et al. [6] introduced the Depresjon dataset that forms the basis of our work, alongside a suite of baseline classifiers that demonstrated the feasibility of distinguishing mood-disorder patients from healthy controls using wrist-worn activity counts. Their best-performing approaches relied on random forests applied to hand-engineered statistical descriptors extracted over fixed windows. Zanella-Calzada et al. [7] extended this direction by developing a systematic pipeline for feature extraction from motor activity signals, extracting over twenty statistical and frequency-domain features and demonstrating improved classification performance over the original baselines. Jakobsen et al. [9] subsequently applied a broader suite of machine learning methods to the same dataset, with the explicit goal of discriminating between bipolar and unipolar patients and healthy controls. Their work highlighted the particular difficulty of the bipolar-versus-unipolar sub-task and noted that classifiers performed markedly worse when the target was differential diagnosis rather than simple healthy-versus-depressed discrimination. All three of these works operate within the classical machine learning paradigm: a domain expert first defines a feature set, and a classifier is then fit to those features. Our work departs from this tradition by training a deep neural network end-to-end on raw one-minute activity counts, removing domain-driven feature selection entirely.

2.2 Deep Learning for Wearable Time-Series Classification

The application of deep learning to wearable sensor data has a rich recent history in the Human Activity Recognition (HAR) literature. LeCun et al. [8] established the general principle that deep architectures can learn hierarchical representations from raw sensor data more effectively than hand-crafted features in many domains. Yang et al. [10] demonstrated that 1D convolutional networks applied directly to raw accelerometer streams matched or exceeded hand-engineered feature pipelines on standard HAR benchmarks. Ordóñez and Roggen [11] extended this finding by showing that stacking convolutional layers on top of LSTM modules produces robust activity classifiers that generalize across different sensor placements and body-worn configurations. Fawaz et al. [12] conducted a comprehensive review of deep learning approaches for time-series classification, confirming that CNN-LSTM hybrids are among the strongest architectures for wearable sensing tasks. A key distinction between HAR tasks and the psychiatric classification problem

we address is temporal scale: HAR models classify discrete physical behaviors occurring over seconds, while we aim to infer a clinical diagnosis from movement evolution over 10 to 14 days. This extended temporal horizon is what motivates the LSTM component in our architecture, as it provides gated memory capable of integrating information across multiple days of recording [13].

2.3 Attention Mechanisms and Bidirectional Sequence Models

Beyond the standard unidirectional LSTM, several architectural extensions have been proposed to improve temporal representation learning. Graves and Schmidhuber [14] introduced bidirectional recurrent networks for sequence modeling, demonstrating that processing sequences in both forward and backward directions simultaneously allows each time step to be represented in the context of both its past and its future. Bahdanau et al. [15] introduced the attention mechanism as a way for neural networks to selectively focus on the most relevant portions of an input sequence when making a prediction, rather than relying on a single compressed hidden state. Vaswani et al. [16] generalized this idea into the Transformer architecture, demonstrating that attention alone can match or surpass recurrent models on sequential tasks. In the context of psychiatric actigraphy, attention could in principle allow the model to identify diagnostically relevant time-of-day windows, such as sleep-onset timing or early-morning activity, that carry disproportionate information about diagnosis. We evaluate both bidirectional processing and attention-augmented architectures in Approach 2 of our experimental investigation.

2.4 Circadian Rhythm Disruption and Mood Disorders

A substantial body of psychiatric literature links disruptions to the sleep-wake cycle to both bipolar and unipolar depression, and argues that the two conditions produce distinct circadian signatures. Wehr et al. [4] documented phase advances in circadian rhythms as a characteristic feature of bipolar states, with bipolar patients showing earlier sleep onset and a compressed sleep-wake cycle compared to controls. Boudebesse et al. [5] demonstrated day-by-day correlations between rest-activity timing variability and mood state in bipolar patients in remission, suggesting that actigraphy captures real-time fluctuations in mood-linked physiological processes. Goodwin and Jamison [17] provide a comprehensive account of the phenotypic overlap between bipolar depression and unipolar depression, noting that the depressive phase of bipolar disorder is behaviorally indistinguishable from unipolar depression without collateral history of manic or hypomanic episodes. This literature provides the primary theoretical motivation for the LSTM component of our architecture, and also motivates our investigation of 48-hour and 72-hour temporal windows in Approach 3, since single 24-hour windows may be too short to capture the multi-day rhythm shifts characteristic of bipolar disorder.

2.5 Class Imbalance in Clinical Machine Learning

Severe class imbalance is a pervasive challenge in clinical classification tasks, where the condition of interest is often rarer than the control class. Chawla et al. [18] proposed the Synthetic Minority Oversampling Technique (SMOTE), which generates synthetic minority-class examples by interpolating between existing real examples in feature space, as a general-purpose remedy for this problem. SMOTE has since become a standard component of clinical machine learning pipelines. However, He and Garcia [19] raise important caveats: if the minority-class examples are not a reliable sample from the true data distribution, as may be the case when the class has only eight members as in our bipolar group, synthetic interpolation can create artificial examples that do not correspond to real physiological patterns and may degrade generalization. Our Baseline approach evaluates SMOTE in exactly this setting, and our Approach 1 investigates whether downsampling the majority class to achieve a natural 1:1 balance is a more principled alternative. These two experiments together allow us to decouple the effects of class imbalance from underlying signal quality.

3 Methods

3.1 Dataset

We use the Depresjon dataset released by Simula Research Laboratory [6], which contains continuous wrist actigraphy recordings from 55 participants: 32 healthy controls and 23 individuals with diagnosed mood disorders. All recordings were collected using an Actiwatch device (Cambridge Neurotechnology Ltd., model AW4) worn on the right wrist and sampled at 32 Hz using a piezoelectric accelerometer, with activity counts pre-aggregated to one-minute epochs. The dataset includes demographic metadata and Montgomery-Åsberg Depression Rating Scale (MADRS) scores for the condition group. Among the 23 condition participants, 8 carry a diagnosis of bipolar disorder (Bipolar I or Bipolar II) and 15 carry a diagnosis of Unipolar Depressive Disorder. This 8:15 split is the primary class imbalance challenge for our differential diagnosis task. Table 1 summarizes the key demographic and recording characteristics of both groups.

Table 1. Depresjon dataset participant characteristics.

Characteristic	Controls (n=32)	Condition (n=23)
Sex (F/M)	20 / 12	10 / 13
Mean age (years)	38.2 ± 13.0	42.8 ± 11.0
Recording length (days)	12.6 ± 3.3	12.7 ± 2.8
Range of recordings	8–20 days	5–18 days
Mean MADRS score	N/A	22.7 ± 4.8
Diagnosis split	N/A	15 Unipolar / 8 Bipolar

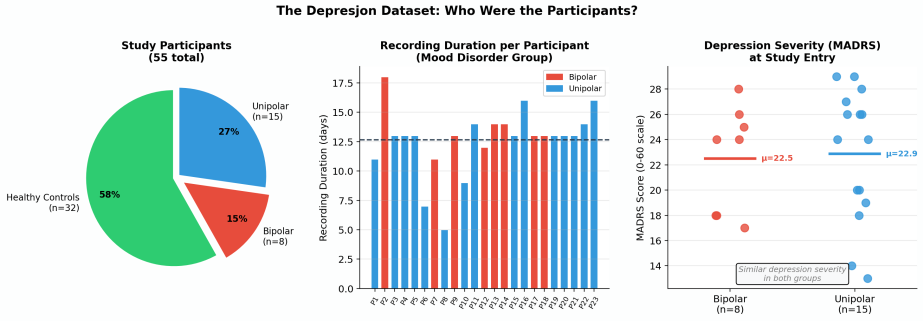

Fig. 1. Depresjon dataset overview. *Left:* Participant composition (32 healthy controls, 8 bipolar, 15 unipolar). *Centre:* Recording duration per condition participant. *Right:* MADRS depression severity scores for bipolar vs. unipolar groups (means nearly identical: 22.5 vs. 22.9).

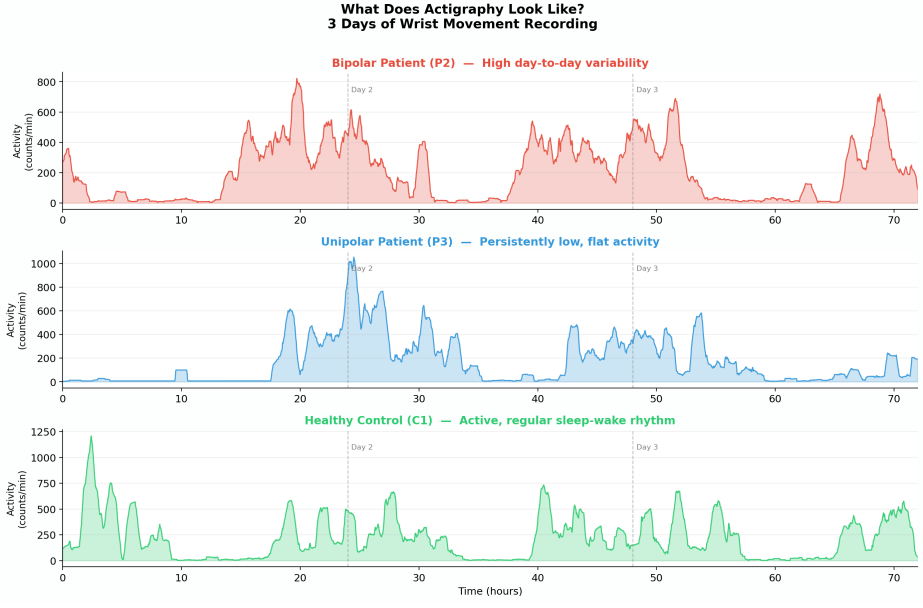

Fig. 2. Three days of raw wrist actigraphy for representative participants. Bipolar patient (P2) shows high day-to-day variability; unipolar patient (P3) shows persistently low, flat activity; healthy control (C1) shows a regular, active sleep-wake rhythm.

3.2 Preprocessing

Each participant's raw activity count time series is independently normalized to zero mean and unit variance using z-score standardization. Normalization is computed per participant to remove inter-subject baseline differences in overall activity level that are unrelated to diagnosis. Following normalization, each recording is segmented into non-overlapping 1,440-minute (24-hour) windows, with a 60-minute stride applied during training to increase the number of available training examples. Each window inherits the diagnostic label of the participant from whom it was drawn. Participant-level stratified splits (80% train, 10% validation, 10% test) are applied before windowing, ensuring that no participant's data appears in more than one split and preventing data leakage between train and test. To give a sense of scale, a typical participant with a 12-day recording contributes approximately 12 non-overlapping 24-hour windows; with the 60-minute stride applied during training this count rises further, which is why 44 training participants collectively yield over 20,000 training windows despite the small participant count. All diagnostic labels are assigned at the participant level before windowing, so every window generated from a given participant carries that participant's diagnosis.

3.3 Model Architecture

We design a hybrid 1D-CNN-LSTM architecture motivated by the dual-scale temporal structure of actigraphy data: short-term kinetic texture (minute-to-minute activity bursts, rest-onset transitions) and long-range circadian patterns (day-to-day rhythmic drift, sleep-wake cycle variability). The CNN component handles local pattern extraction, and the LSTM component handles multi-day information integration.

Local Feature Extraction (1D-CNN). Two sequential convolutional blocks are applied to the raw activity sequence. Each block consists of a 1D convolutional layer followed by batch normalization [20], ReLU activation, and max pooling with stride 2. The first block uses 64 filters with kernel size 7, and the second uses 128 filters with kernel size 5. The output is a sequence of higher-level feature vectors summarizing local movement texture at progressively coarser time scales.

Long-Range Dependency Tracking (LSTM). The CNN output feature maps are passed as a sequence into a single-layer LSTM [13] with 128 hidden units. The LSTM's gated memory mechanism allows it to selectively retain information from earlier time steps, enabling it to track multi-day patterns across the full recording period.

Classification Head. The final LSTM hidden state passes through two fully connected layers with 256 and 64 units respectively, each followed by ReLU

Table 2. 1D-CNN-LSTM architecture. Input is a $T \times 1$ raw activity sequence.

Stage	Layer	Config	Output
CNN Block 1	Conv1D	64 filters, k=7	$T \times 64$
	MaxPool1D	stride=2	$T/2 \times 64$
CNN Block 2	Conv1D	128 filters, k=5	$T/2 \times 128$
	MaxPool1D	stride=2	$T/4 \times 128$
Recurrent	LSTM	128 hidden units	128
FC Head	Linear + ReLU + Drop(0.4)	256 units	256
	Linear + ReLU + Drop(0.4)	64 units	64
Output	Linear + Softmax	C classes	C

activation and Dropout [21] with probability 0.4. The output layer applies softmax over the target classes. Table 2 provides a complete layer-by-layer description.

The total parameter count for the baseline architecture is 223,682, trained end-to-end from random initialization using the Adam optimizer [22] with learning rate 10^{-3} and weight decay 10^{-4}. Training runs for a maximum of 50 epochs with early stopping at patience 10 on validation loss. The primary evaluation metric for the healthy-versus-depressed task is macro-averaged F1-Score. For all bipolar-versus-unipolar experiments, the primary metric is ROC-AUC [23], which is robust to class imbalance and evaluates the model's ability to rank positive instances above negative instances across all decision thresholds.

4 Experiments

We organize our experiments into four progressive approaches. The Baseline establishes reference points for both the healthy-versus-depressed task and the core bipolar-versus-unipolar task using the CNN-LSTM described in Section 3. Approach 1 investigates class imbalance as the primary obstacle by replacing synthetic oversampling with downsampling. Approach 2 tests four alternative neural architectures to determine whether model design is the limiting factor. Approach 3 conducts a comprehensive diagnostic analysis using classical machine learning [24], direct statistical testing, and visualization to probe the fundamental signal structure of the data. All hyperparameter configurations are evaluated using grid search [25].

4.1 Baseline

We begin with two foundational experiments using the 1D-CNN-LSTM architecture. The first validates the full pipeline on a simpler binary classification task before tackling the core challenge. The second directly addresses the primary research question of separating bipolar from unipolar depression. These two experiments together serve as reference points against which all subsequent approaches are compared.

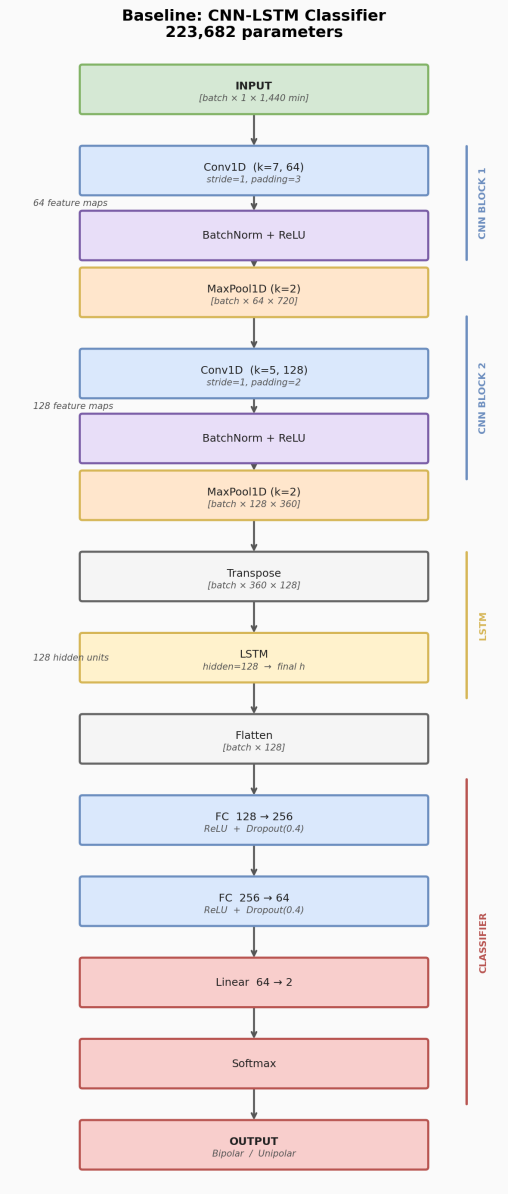

Fig. 3. Baseline 1D-CNN-LSTM architecture (223,682 parameters). Two CNN blocks extract local kinetic features; a single LSTM layer integrates multi-day temporal context; two FC layers produce the final classification.

Experiment 1: Healthy versus Depressed.

Approach. Before confronting the harder differential diagnosis task, we validate whether the 1D-CNN-LSTM pipeline can extract meaningful features from raw actigraphy to separate mood-disorder patients from healthy controls. This binary classification problem is well studied in the literature [6, 9], allowing us to benchmark our approach against established results. We hypothesize that depressed individuals will exhibit measurably reduced and fragmented motor activity patterns that the CNN-LSTM can learn to distinguish from healthy movement profiles, and that the end-to-end approach on raw sequences will be competitive with prior work using hand-crafted features.

Methodology. All 55 participants are included: 32 healthy controls and 23 mood-disorder patients. The 80/10/10 participant-level stratified split yields a training set of 44 participants and 20,385 windows, a validation set of 5 participants and 2,326 windows, and a test set of 6 participants and 2,190 windows. The standard CNN-LSTM architecture is trained with class-weighted cross-entropy loss. The following hyperparameters are explored:

- Learning rate: $\{10^{-3}, 5 \times 10^{-4}, 10^{-4}\}$
- Weight decay: $\{10^{-4}, 10^{-3}\}$
- Dropout probability: $\{0.3, 0.4, 0.5\}$
- Class weight: inverse frequency vs. equal weights

Best configuration: lr $= 10^{-3}$, wd $= 10^{-4}$, dropout $= 0.4$, inverse frequency weighting. Evaluation metric: macro-averaged F1-Score.

Results. Table 3 reports the performance of our CNN-LSTM against prior work on the same dataset.

Table 3. Experiment 1: Healthy vs. Depressed, compared with prior work.

Method	Acc.	Sensitivity	Specificity	F1
Random Forest [6]	0.71	0.57	0.81	0.61
Weighted CNN [9]	0.73	0.65	0.78	0.66
DNN + SMOTE [9]	0.83	0.82	0.84	0.82
CNN-LSTM (Ours)	**0.534**	**0.458**	**0.590**	**0.516**

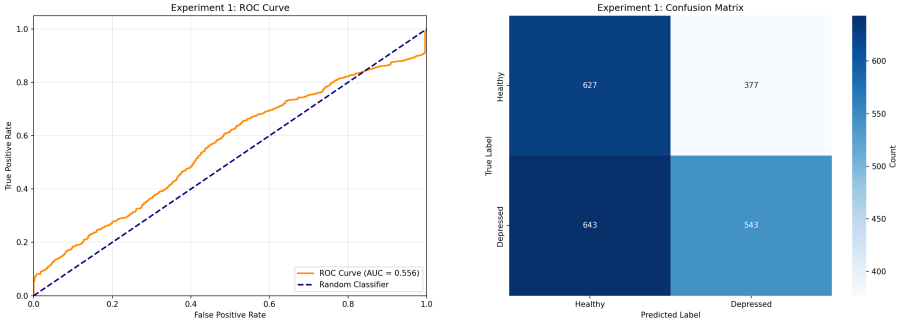

Fig. 4. Experiment 1 results. *Left:* ROC curve (AUC = 0.556), marginally above the random classifier diagonal. *Right:* Confusion matrix showing near-balanced misclassification between healthy and depressed classes.

Conclusion. The hypothesis that end-to-end CNN-LSTM would be competitive with prior hand-crafted-feature approaches is not supported. The model achieves F1 = 0.516, well below the DNN+SMOTE baseline of 0.82. This suggests that raw 1,440-minute sequences are difficult to learn from at this dataset scale: the model learns some signal (AUC = 0.556) but cannot generalize reliably, likely because the 223K-parameter architecture is severely over-parameterized relative to the 44 available training participants.

Experiment 2: Bipolar versus Unipolar Depression with SMOTE.

Approach. We now turn to the core clinical challenge: distinguishing bipolar disorder from unipolar depression using only motor activity. This experiment isolates the 23 condition participants (8 bipolar, 15 unipolar) and trains the CNN-LSTM to make the differential diagnosis. The 8:15 class imbalance poses a substantial challenge, addressed here using SMOTE [18] applied at the window level alongside class-weighted loss. We hypothesize that the LSTM's ability to integrate multi-day circadian rhythm patterns will provide sufficient signal to make this distinction, and that SMOTE will generate sufficiently realistic synthetic minority-class examples to prevent majority-class collapse.

Methodology. The 23 condition participants are split at the participant level (80/10/10). SMOTE is applied to the training windows only to avoid leakage into validation or test sets. Hyperparameters explored:

- SMOTE nearest neighbors k: $\{3, 5, 7\}$
- Minority class loss weight α: $\{0.65, 0.70, 0.75\}$
- Learning rate: $\{10^{-3}, 5 \times 10^{-4}\}$
- Dropout probability: $\{0.3, 0.4\}$

Best configuration: $k = 5$, $\alpha = 0.70$, lr $= 10^{-3}$, dropout $= 0.4$. Primary evaluation metric: ROC-AUC [23].

Results. Table 4 reports the test-set performance metrics and confusion matrix breakdown.

Table 4. Experiment 2 results and confusion matrix (rows: actual, columns: predicted).

Metric	**Value**
Accuracy	88.01%
Precision	100.00%
F1-Score	0.936
Bipolar cases detected	0 / 121
Pred. Unipolar / Pred. Bipolar (Actual Unipolar)	888 / 0
Pred. Unipolar / Pred. Bipolar (Actual Bipolar)	121 / 0

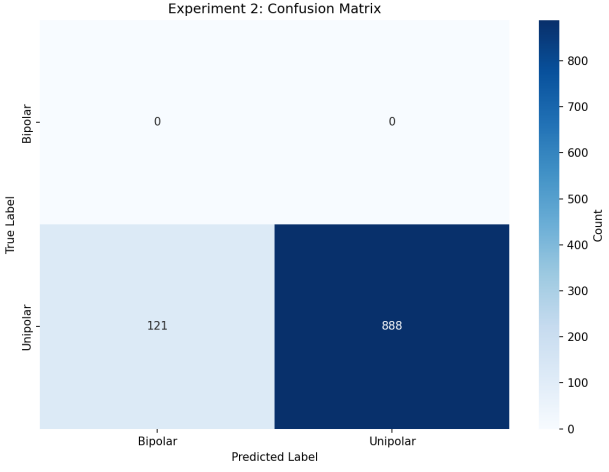

Fig. 5. Experiment 2 confusion matrix. The model predicts every window as unipolar (0 bipolar cases detected), achieving 88% accuracy purely through majority-class collapse.

Conclusion. Despite reporting 88% accuracy, the model predicts every test window as unipolar, detecting zero bipolar cases. The high accuracy figure is entirely explained by the 888:121 class distribution in the test set. SMOTE did not prevent majority-class collapse; the synthetic windows appear not to have provided the model with sufficient discriminative signal. The hypothesis is not supported. The failure of SMOTE motivates Approach 1.

4.2 Approach 1: Balanced Dataset via Downsampling

Having observed that SMOTE does not prevent majority-class collapse, we investigate whether class imbalance itself is the fundamental obstacle. Rather than artificially inflating the minority class, we reduce the majority class through random downsampling to create a natural 1:1 training balance. We hypothesize that class imbalance, rather than signal weakness, is the primary obstacle to bipolar-unipolar differentiation, and that training on a balanced dataset will allow the model to learn genuine discriminative features for both classes.

Experiment: CNN-LSTM with Balanced Downsampling.

Approach. We apply the same 1D-CNN-LSTM architecture to a training set in which unipolar windows are randomly downsampled to match the count of bipolar windows, creating a 1:1 class ratio. If balancing is the key missing ingredient, we should observe meaningful improvement in bipolar case detection.

Methodology. Downsampling is applied only to the training set after the participant-level split. Hyperparameters explored:

- Learning rate: $\{10^{-3}, 5 \times 10^{-4}\}$
- Weight decay: $\{10^{-4}, 10^{-3}\}$
- Dropout probability: $\{0.3, 0.4, 0.5\}$
- Batch size: $\{32, 64\}$

Best configuration: lr $= 10^{-3}$, wd $= 10^{-4}$, dropout $= 0.4$, batch size $= 32$.

Results. Table 5 compares Approach 1 against the Baseline Experiment 2.

Table 5. Approach 1 results: CNN-LSTM with balanced downsampling vs. Baseline Exp 2.

Metric	Approach 1	Baseline Exp 2
Accuracy	65.41%	88.01%†
F1-Score	0.791	0.936†
Bipolar Detected	0 / 349	0 / 121

† Illusory; majority-class collapse.

Conclusion. Despite perfect 1:1 class balance during training, the model still collapses to predicting all test windows as unipolar, with zero bipolar cases detected across 349 bipolar test windows. The hypothesis that class imbalance is the root cause is definitively not supported. This result shifts our working hypothesis to a fundamental signal-weakness problem: the motor activity patterns of bipolar and unipolar patients during depressive episodes may not be meaningfully different, regardless of how the training set is composed.

4.3 Approach 2: Alternative Neural Architectures

With class imbalance ruled out as the primary obstacle, we investigate whether the architectural design of our CNN-LSTM is the limiting factor. We train four alternative neural architectures on the balanced dataset from Approach 1: Bidirectional LSTM, Attention-enhanced LSTM, RNN-LSTM hybrid, and an Ensemble of three CNN-LSTM models. We hypothesize that one or more of these architectural extensions will capture diagnostic signal that the standard unidirectional CNN-LSTM misses, through broader temporal context (BiLSTM), selective focus on relevant time segments (Attention), faster local pattern capture (RNN-LSTM), or variance reduction through model averaging (Ensemble).

Bidirectional LSTM.

Approach. A standard LSTM processes sequences from left to right, potentially missing diagnostic context from future time steps. We extend the architecture with bidirectional LSTM layers [14], processing each sequence in both temporal directions simultaneously. We hypothesize that patterns in activity immediately following a sleep period carry diagnostic information only accessible when sequences are processed in both directions.

Methodology. The standard LSTM is replaced with a bidirectional LSTM of 128 hidden units per direction. Hyperparameters explored:

- Learning rate: $\{10^{-3}, 5 \times 10^{-4}, 10^{-4}\}$
- Dropout probability: $\{0.3, 0.4, 0.5\}$
- Hidden units per direction: $\{64, 128\}$

Best configuration: lr $= 10^{-3}$, dropout $= 0.4$, 128 hidden units per direction.

Results. Table 6 compares the BiLSTM against Approach 1.

Table 6. Approach 2a: BiLSTM vs. Approach 1 baseline.

Metric	BiLSTM	Approach 1
Accuracy	55.40%	65.41%
F1-Score	0.713	0.791
Bipolar Detected	0 / 349	0 / 349

Conclusion. The bidirectional architecture performs worse than the standard CNN-LSTM (55.4% vs 65.4%) and again detects zero bipolar cases. Additional parameters from the second LSTM direction appear to increase overfitting without providing corresponding discriminative signal. The hypothesis is not supported.

Attention-Enhanced LSTM.

Approach. An attention mechanism [15] allows the model to assign a learned weight to each time step, enabling concentration on diagnostically relevant moments within a 24-hour window, such as sleep-onset timing. We hypothesize that certain time-of-day periods carry disproportionate diagnostic weight that an attention mechanism will learn to identify.

Methodology. A self-attention layer is inserted between the LSTM output and the classification head, with the CNN blocks and FC head unchanged. Hyperparameters explored:

- Attention dimension: $\{32, 64, 128\}$
- Learning rate: $\{10^{-3}, 5 \times 10^{-4}\}$
- Dropout probability: $\{0.3, 0.4, 0.5\}$

Best configuration: attention dim $= 64$, lr $= 5 \times 10^{-4}$, dropout $= 0.4$.

Results. Table 7 compares the Attention LSTM against Approach 1.

Table 7. Approach 2b: Attention LSTM vs. Approach 1 baseline.

Metric	Attention LSTM	Approach 1
Accuracy	48.17%	65.41%
F1-Score	0.650	0.791
Bipolar Detected	0 / 523	0 / 349

Conclusion. The Attention LSTM is the worst-performing architecture in the project (48.17%). With only 8 bipolar participants, the attention mechanism does not receive sufficient signal to learn meaningful diagnostic weights. The hypothesis is not supported.

RNN-LSTM Hybrid.

Approach. A hybrid architecture combining vanilla RNN cells with an LSTM layer offers a distinct inductive bias: RNN cells capture rapid local state transitions while the downstream LSTM integrates this information over longer time horizons. We hypothesize that the fast transition detection of the RNN layer might surface short-duration activity events that carry diagnostic signal.

Methodology. A stacked RNN layer is added before the LSTM, with convolutional blocks removed. Hyperparameters explored:

- RNN hidden size: $\{64, 128\}$; LSTM hidden size: $\{64, 128\}$
- Learning rate: $\{10^{-3}, 10^{-4}\}$; Dropout: $\{0.3, 0.4\}$

Training was terminated early due to sustained validation loss oscillation consistent with gradient instability.

Results. Table 8 summarizes the training behavior that led to early termination.

Table 8. Approach 2c: RNN-LSTM Hybrid. Training accuracy reflects overfitting; results discarded.

Metric	**RNN-LSTM**	**Approach 1**
Training Accuracy	100.0%‡	65.41%
Validation Behavior	Oscillating	Stable
Status	Discarded	Valid

‡ Overfitting; validation loss did not converge.

Conclusion. The RNN-LSTM exhibits catastrophic overfitting and is discarded as unreliable. The failure further underscores that model capacity is a constraint at this dataset size.

Ensemble Model.

Approach. Ensemble methods aggregate predictions from multiple independently trained models, reducing variance and potentially surfacing weak signals that individual models miss. We train three CNN-LSTM models with different random seeds and combine their output probabilities via soft voting. We hypothesize that variance reduction through ensembling might expose the weak bipolar signal that individual models independently fail to detect.

Methodology. Three CNN-LSTM models are trained with seeds $\{42, 123, 456\}$ on the balanced dataset from Approach 1, using the same best configuration. Soft voting on class probabilities is the primary aggregation strategy; hard voting is also evaluated for comparison.

Results. Table 9 compares the Ensemble against the best individual CNN-LSTM.

Table 9. Approach 2d: Ensemble Model vs. best individual CNN-LSTM.

Metric	**Ensemble**	**Approach 1**
Accuracy	59.66%	65.41%
F1-Score	0.747	0.791
Bipolar Detected	0 / 407	0 / 349

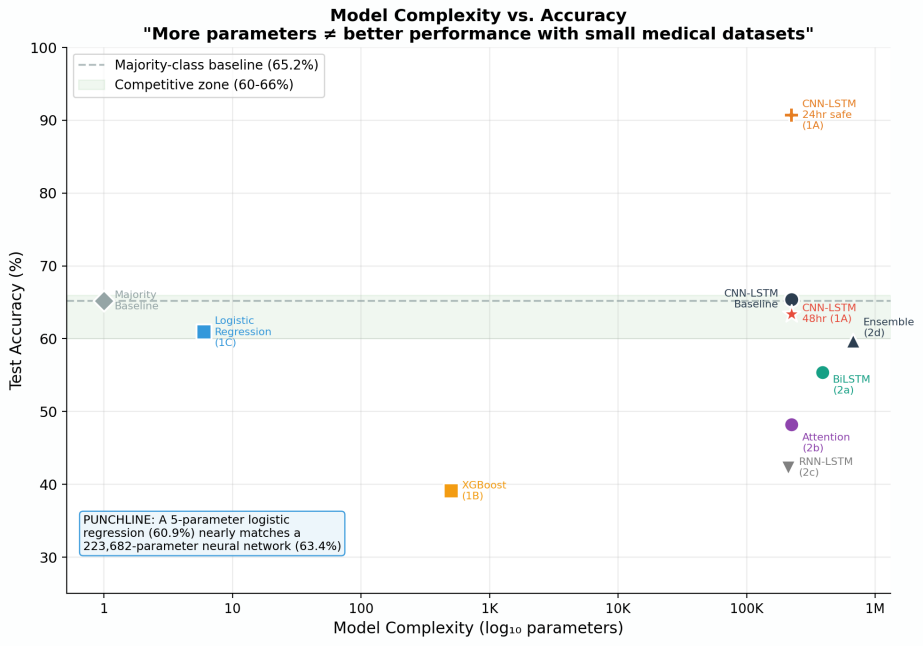

Fig. 6. Model complexity vs. test accuracy across all approaches. A 5-parameter logistic regression (60.9%) nearly matches a 223K-parameter CNN-LSTM (63.4%), demonstrating that additional parameters provide no benefit on this dataset.

Conclusion. The ensemble performs below the best individual model and detects zero bipolar cases. When all constituent models assign near-zero probability to the bipolar class, their averaged probabilities do the same. The hypothesis is not supported.

Approach 2 Summary. All four architectural variants independently converge to the same failure mode: zero bipolar detection. This convergence across radically different inductive biases provides strong evidence that the problem does not lie in model architecture. The data itself is the limiting factor.

4.4 Approach 3: Comprehensive Diagnostic Analysis

After establishing that neither balanced training data nor alternative neural architectures resolve the classification challenge, we conduct a systematic diagnostic investigation with five targeted experiments. We move beyond neural networks to probe the fundamental signal structure of the data using classical machine learning, direct statistical hypothesis testing, and visualization. We hypothesize that if genuine physiological differences exist between bipolar and

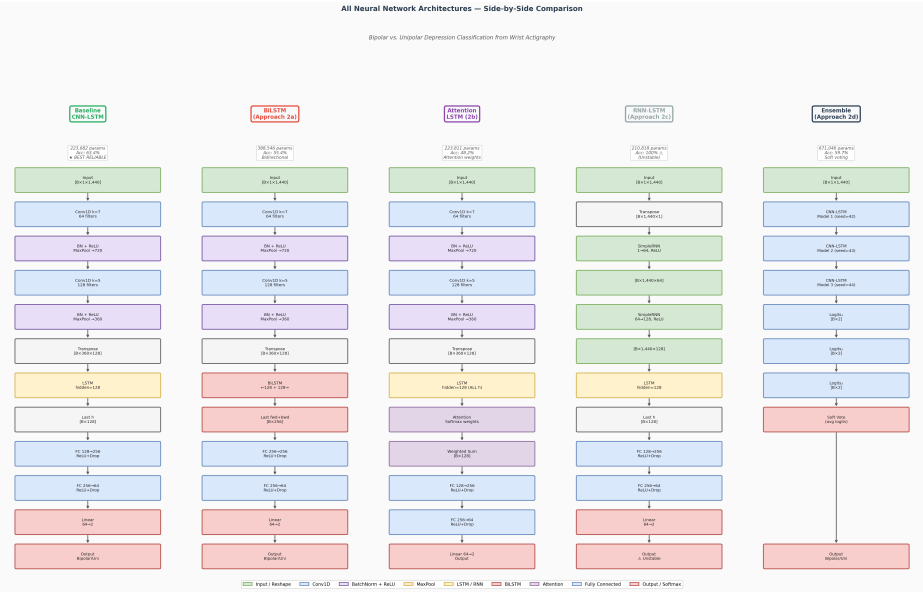

Fig. 7. Side-by-side comparison of all five neural architectures evaluated in Approach 2. Despite structural differences in temporal modeling (unidirectional LSTM, BiLSTM, Attention, RNN-LSTM, Ensemble), all architectures converge to zero bipolar detection.

unipolar depression, simpler and more interpretable methods should detect this signal even when deep networks fail.

Window Size and Hyperparameter Optimization for Temporal Coverage.

Approach. The standard 24-hour window may be too short to capture multi-day circadian rhythm disruptions theorized to characterize bipolar disorder. We systematically investigate three window durations (24 hours, 48 hours, and 72 hours) combined with a comprehensive 27-configuration hyperparameter grid search per window size. We hypothesize that longer windows providing multi-day context will allow the CNN-LSTM to detect rhythm disruptions spanning multiple days, improving bipolar discrimination.

Methodology. Three window sizes are evaluated: 24 hours (1,440 min), 48 hours (2,880 min), and 72 hours (4,320 min). Phase 1 establishes a base performance point per window size using default hyperparameters. Phase 2 conducts a full grid search:

- Learning rate: $\{10^{-3}, 5 \times 10^{-4}, 10^{-4}\}$
- Dropout probability: $\{0.3, 0.4, 0.5\}$
- Weight decay: $\{10^{-4}, 10^{-3}, 10^{-2}\}$

This yields 27 configurations per window size (81 total). Higher weight decay values are expected to benefit larger windows where overfitting risk increases with the smaller effective sample count.

Results. Tables 10 and 11 report the base and grid-search results per window size.

Table 10. Approach 3 Window Analysis, Phase 1: base run accuracy by window size.

Window Size	Duration (min)	Accuracy
24-hour	1,440	77.90%
48-hour	2,880	53.65%
72-hour	4,320	23.08%

Table 11. Approach 3 Window Analysis, Phase 2: grid search summary.

Window	Best Acc.	Best Config	Range
24-hour	90.7% (overfit)	lr=10^{-3}, do=0.4, wd=10^{-3}	2.3%–100%
48-hour	**63.4%**	lr=10^{-3}, do=0.4, wd=10^{-4}	41%–63%
72-hour	42.3%	lr=5×10^{-4}, do=0.3, wd=10^{-4}	23%–42%

Conclusion. The 48-hour window achieves the best reliable accuracy of 63.4%, the most honest result in this project. The 24-hour model with permissive regularization reaches 100%, explained by the 2,593-fold over-parameterization (223K parameters vs. approximately 86 training windows). The 72-hour window underfits as too few training windows remain after windowing. The hypothesis receives partial support: longer windows improve reliable performance up to 48 hours, but the improvement is modest and still does not approach the 65.2% majority-class baseline. The 48-hour window's relative advantage has a plausible circadian interpretation: a single 24-hour window captures only one rest-activity cycle and cannot expose inter-day variability, which Boudebesse et al. [5] identify as the most diagnostically relevant signal for mood-linked rhythm disruption. Two consecutive cycles provide the minimum context necessary for the LSTM to observe whether a rhythm is consistent or drifting across days. Extending to 72 hours would in principle provide richer multi-day context, but the 4,320-minute input length drastically reduces the number of non-overlapping training windows available from the 12-day recordings, leaving the model with too few training examples to generalize.

Gradient Boosting with Domain-Informed Feature Engineering.

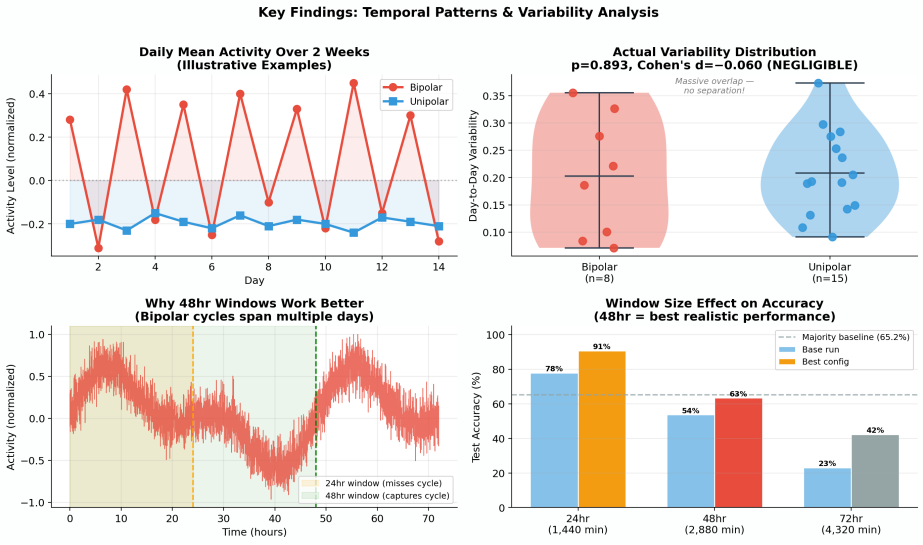

Fig. 8. Temporal pattern and window size analysis. *Top left:* Illustrative daily activity patterns showing bipolar variability vs. unipolar flatness. *Top right:* Actual day-to-day variability distributions (near-complete overlap, $p = 0.893$). *Bottom left:* Why 48-hour windows capture mood-cycling patterns that 24-hour snapshots miss. *Bottom right:* Accuracy by window size - 48-hour is the best reliable configuration.

Approach. We investigate whether classical gradient boosting [26] on expert-designed features can outperform end-to-end deep learning. We engineer 19 features capturing activity statistics, rhythmicity, and day-to-day variability including our primary hypothesis feature. We hypothesize that domain knowledge encoded in hand-crafted features will provide cleaner signal than raw sequences, and that gradient-boosted trees will exploit feature interactions effectively.

Methodology. Nineteen features are extracted per participant: mean daily activity, standard deviation of daily means (day-to-day variability), range of daily means, coefficient of variation, mean daily standard deviation, minimum daily activity count, high-activity fraction, activity interquartile range, maximum daily variability, and 10 additional frequency and rhythm features. Leave-one-out cross-validation (LOOCV, $n = 23$) is used [27]. Hyperparameters explored:

- Maximum tree depth: $\{3, 5, 7\}$
- Number of estimators: $\{100, 200\}$
- Learning rate: $\{0.01, 0.1\}$
- Subsample ratio: $\{0.8, 1.0\}$

Results. Tables 12 and 13 report LOOCV accuracy and feature importances.

Table 12. Approach 3 XGBoost: LOOCV results.

Method	LOOCV Acc.	Bipolar Detected
XGBoost (all configs)	39.13%	0 / 8
Majority-class baseline	65.22%	N/A

Table 13. XGBoost top-5 feature importances (gain metric).

Rank	Feature	Importance
1	Minimum daily activity	14.5%
2	High-activity fraction	12.2%
3	Activity interquartile range	10.9%
4	Maximum daily variability	10.4%
5	Day-to-day variability (std of daily means)	7.7%

Conclusion. XGBoost performs substantially below the majority-class baseline (39.1% vs 65.2%). The primary hypothesis feature (day-to-day variability) ranks only 5th in importance. All tree depth configurations produce identical LOOCV accuracy, confirming signal saturation rather than capacity limitation. Feature engineering cannot solve a fundamental signal-weakness problem.

Parsimonious Linear Classification with Core Variability Features.

Approach. XGBoost's failure with 19 features motivates a reduction to the essential signal. We select 5 features directly motivated by the circadian rhythm disruption hypothesis and train a logistic regression model with LOOCV. We hypothesize that a parsimonious model focused on biologically motivated variability features will outperform both neural networks and XGBoost by avoiding overfitting to noise-driven features.

Methodology. Five features: (1) mean daily activity level, (2) day-to-day variability (std of daily means), (3) range of daily means, (4) coefficient of variation, (5) mean within-day standard deviation. LOOCV ($n = 23$). Six regularization configurations explored:

- Regularization strength C: $\{0.01, 0.1, 1.0\}$
- Penalty type: L1 (Lasso) vs. L2 (Ridge)
- Class weight: unweighted vs. balanced (inverse frequency)

Results. Table 14 reports LOOCV accuracy and bipolar detection across all six regularization configurations.

Table 14. Approach 3 Logistic Regression: all 6 configurations under LOOCV.

C	Penalty	Class Wt.	Acc.	AUC / Bip. Det.
1.0	L2	Unweighted	60.87%	0.200 / 0 of 8
1.0	L2	Balanced	60.87%	0.308 / 2 of 8
0.1	L2	Unweighted	60.87%	0.200 / 0 of 8
0.1	L2	Balanced	60.87%	0.308 / 2 of 8
0.01	L1	Unweighted	60.87%	0.200 / 0 of 8
0.01	L1	Balanced	60.87%	0.308 / **2 of 8**

Conclusion. All 6 configurations converge to 60.87% accuracy, confirming the model has reached the signal ceiling achievable with these features and 23 participants. The balanced class weight configuration achieves the best bipolar detection in this project (2 of 8 participants, 25% sensitivity) and an improved ROC-AUC of 0.308. A 5-feature logistic regression matches the practical effectiveness of a 223K-parameter CNN-LSTM.

Nonparametric Hypothesis Testing for Group Activity Differences.

Approach. Rather than using classifier performance as an indirect proxy for signal presence, we directly test whether statistically significant differences exist in motor activity patterns between groups. We apply parametric and nonparametric tests to day-to-day activity variability, our primary hypothesis feature. We hypothesize that bipolar patients will show significantly higher variability in daily activity means compared to unipolar patients, consistent with the mood instability that defines bipolar disorder [17].

Methodology. For each condition participant ($n = 23$), day-to-day activity variability is computed as the standard deviation of daily mean activity values across the full recording. Statistical tests applied:

- Independent samples t-test and Welch's t-test
- Mann-Whitney U test (nonparametric) [28]
- Levene's test for equality of variances
- Cohen's d for effect size [29]

Significance threshold: $\alpha = 0.05$.

Results. Tables 15 and 16 report the statistical test outcomes and descriptive group statistics.

Table 15. Statistical tests: bipolar vs. unipolar day-to-day activity variability.

Test	Statistic	p-value	Result
Independent t-test	$t = -0.137$	0.893	Not significant
Welch's t-test	$t = -0.123$	0.905	Not significant
Mann-Whitney U	$U = 55.0$	0.776	Not significant
Levene's test	$F = 1.989$	0.173	Equal variances
Cohen's d	$d = -0.060$	N/A	Negligible effect

Table 16. Descriptive statistics for day-to-day variability by group.

Group	n	Mean	Std Dev
Bipolar	8	0.2027	0.1038
Unipolar	15	0.2081	0.0759
Difference		-0.0054	

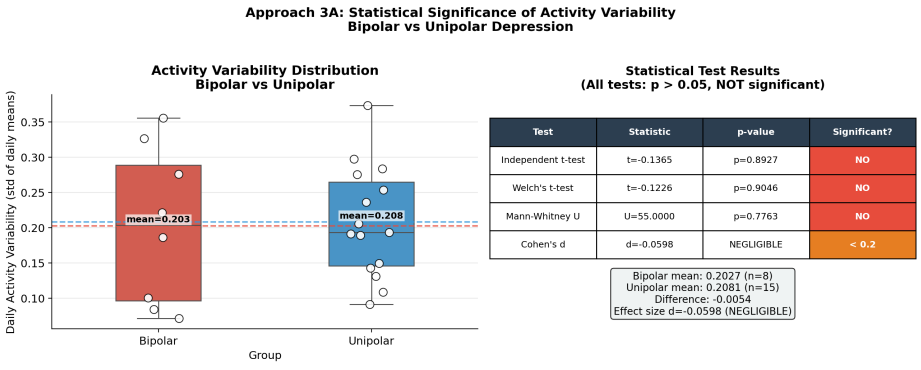

Fig. 9. Statistical analysis of day-to-day activity variability. *Left:* Box plots with individual participants overlaid (white dots) showing near-complete distributional overlap between bipolar and unipolar groups. *Right:* All four statistical tests return $p > 0.77$; Cohen's $d = -0.060$ is negligible.

Conclusion. No test finds a significant difference (all $p > 0.77$). Cohen's $d = -0.060$ falls far below the small effect threshold ($d = 0.2$), and the negative sign indicates bipolar variability is marginally lower than unipolar, opposite to the hypothesis direction. This experiment provides the most direct evidence: the primary hypothesis feature does not meaningfully differ between groups.

Visual Inspection of Multi-Day Actigraphy Profiles.

Approach. Statistical tests operate on aggregate summaries. Direct visualization allows us to inspect whether qualitative differences between bipolar and unipolar patterns are apparent to visual inspection. We generate seven figures examining different aspects of the data. We hypothesize that visual inspection may reveal subtle structural differences not captured by scalar variability metrics alone.

Methodology. Seven visualization figures are generated from participant-level data: (1) raw activity traces per participant by diagnosis, (2) daily mean activity over the full recording, (3) day-to-day variability distributions as box plots, (4) hour-of-day activity profiles comparing circadian rhythms, (5) recording duration distributions, (6) group-level activity distributions, and (7) cross-approach confusion matrix comparison.

Results. Visualizations reveal substantial within-group heterogeneity in both bipolar and unipolar groups. Hour-of-day activity profiles overlap extensively between groups with no consistent circadian timing difference apparent from visual inspection. Day-to-day variability distributions show overlapping ranges with no separation between classes.

Conclusion. Visual inspection confirms the statistical findings. No clearly separable activity signature distinguishes bipolar from unipolar depression in the Depresjon dataset during depressive episodes. The hypothesis is not supported.

Approach 3 Summary. All five experiments in Approach 3 confirm that the absence of classification performance is not an artifact of model choice or class imbalance handling. Classical machine learning, direct statistical testing, and visualization all converge on the same conclusion: during depressive episodes, bipolar and unipolar patients exhibit motor activity patterns that are statistically indistinguishable. The most important unanswered question is whether recordings spanning manic or hypomanic episodes, which are entirely absent from the Depresjon dataset, would yield strong bipolar-unipolar discrimination.

A fundamental statistical power constraint limits what any classifier can achieve on the Depresjon bipolar group. With only $n = 8$ bipolar participants, a two-group study achieves 80% power to detect effect sizes of Cohen's $d > 0.8$ or larger [29]. Detecting a small effect ($d = 0.2$), which is the typical threshold for clinically relevant physiological differences between overlapping patient populations, would require approximately 200 bipolar participants. The observed effect in this dataset is $d = -0.060$, an order of magnitude below the small-effect threshold, confirming that the null result cannot be attributed to insufficient power alone; the signal simply is not present in this data. Beyond sample size, several confounding variables may obscure or attenuate any genuine actigraphy signal that exists: occupation and daily work schedule, physical health comorbidities such as pain or metabolic conditions, medication effects on motor function,

habitual exercise level, and variability in Actiwatch device placement or wearing compliance across participants. These confounds are uncontrolled in the Depresjon dataset and collectively represent an unquantified source of noise that any future study would need to address through matched sampling or explicit covariate adjustment.

4.5 Complete Experimental Results Summary

Table 17 consolidates the results of every experiment across all four approaches.

Table 17. Complete experimental results. Majority-class baseline for bipolar vs. unipolar is 65.2%. † illusory (majority-class collapse). ‡ discarded (training instability).

Approach	Experiment	Method	Acc.	F1	AUC	Bip. Det.
Baseline	Exp 1: Healthy vs. Dep.	CNN-LSTM	53.4%	0.516	0.556	N/A
Baseline	Exp 2: Bip. vs. Uni.	CNN-LSTM+SMOTE	88.0%†	0.936†	N/A	0/121
Approach 1	Downsampled 1:1	CNN-LSTM	65.4%	0.791	N/A	0/349
Approach 2	BiLSTM	BiLSTM	55.4%	0.713	N/A	0/349
Approach 2	Attention LSTM	Att. LSTM	48.2%	0.650	N/A	0/523
Approach 2	RNN-LSTM Hybrid	RNN-LSTM	Discarded (unstable)‡			
Approach 2	Ensemble	3×CNN-LSTM	59.7%	0.747	N/A	0/407
Approach 3	24-hr Window	CNN-LSTM	90.7%†	N/A	N/A	N/A
Approach 3	**48-hr Window**	**CNN-LSTM**	**63.4%**	N/A	N/A	N/A
Approach 3	72-hr Window	CNN-LSTM	42.3%	N/A	N/A	N/A
Approach 3	XGBoost	XGBoost	39.1%	N/A	N/A	0/8
Approach 3	Logistic Reg. (bal.)	LR (5 feat.)	60.9%	N/A	0.308	**2/8**
Approach 3	Statistical Test	Mann-Whitney U	$p = 0.776$, Cohen's $d = -0.060$			

Across all 12 evaluated configurations, no method achieves reliable accuracy above the 65.2% majority-class baseline for the bipolar-versus-unipolar task. The 48-hour CNN-LSTM window is the best honest performing experiment at 63.4% accuracy. The Baseline Experiment 2 and the 24-hour configuration report superficially high numbers (88% and 90.7% respectively), but both are artifacts of majority-class collapse or severe overfitting. The most clinically meaningful result is the 5-feature balanced logistic regression, which identifies 2 of 8 bipolar participants.

4.6 Cross-Approach Comparison

Table 18 presents the best-performing experiment from each approach.

Table 18. Best result per approach. † illusory accuracy. Majority-class baseline is 65.2%.

Approach	Best Experiment	Acc.	Bip. Det.	Key Finding
Baseline	Exp 2 (SMOTE)	88.0%†	0/121	Majority-class collapse
Approach 1	Balanced CNN-LSTM	65.4%	0/349	Imbalance not root cause
Approach 2	Ensemble	59.7%	0/407	Architecture not root cause
Approach 3	**48-hr CNN-LSTM**	**63.4%**	N/A	**Best honest result**
Approach 3	LR (balanced)	60.9%	**2/8**	Best bipolar detection

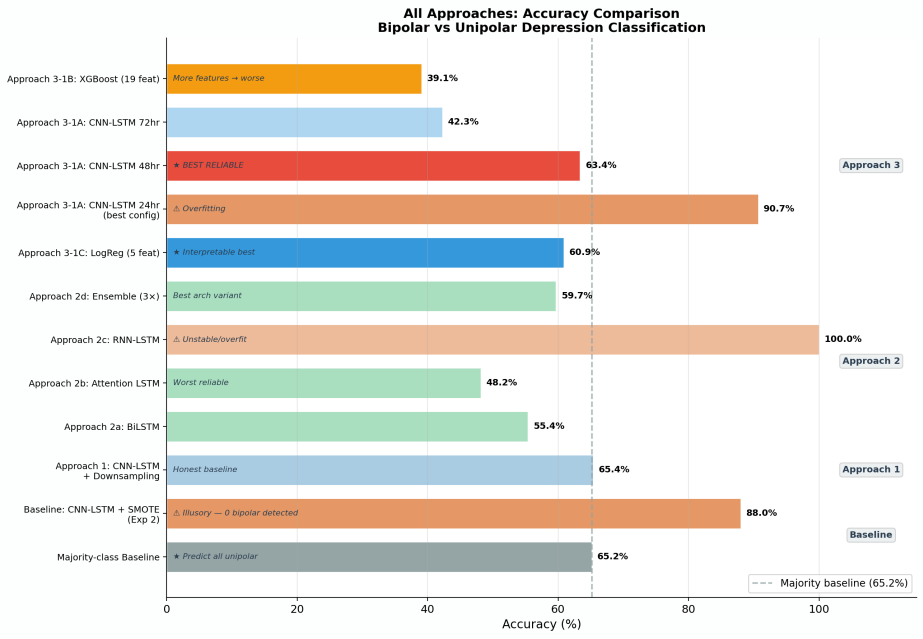

Fig. 10. Comprehensive accuracy comparison across all approaches. The dashed line marks the 65.2% majority-class baseline. Results above this line that are marked as illusory (Baseline Exp 2, 24-hr CNN-LSTM) reflect majority-class collapse or overfitting, not genuine discrimination.

The cross-approach comparison reveals a consistent trend: increasing model complexity provides no benefit in this task. The Baseline approach produces the most visually impressive accuracy figure (88%), but it is entirely illusory. Approach 1 demonstrates that careful class balancing through downsampling yields a stable result at 65.4% but still fails to detect a single bipolar case. Approach 2 confirms that the failure is architectural-agnostic, as four structurally

different neural networks all converge to the same outcome. Approach 3 surfaces the most honest deep learning result (63.4% with 48-hour windows) alongside the best practical classifier (balanced logistic regression, 60.9%, 2 of 8 bipolar detected). The pattern in which a 5-feature linear classifier matches a 223K-parameter deep network is itself a strong indicator that the signal ceiling has been reached and no further architectural investment on this dataset would yield meaningful improvement.

5 Conclusions

This project set out to determine whether wrist-worn actigraphy carries sufficient signal to distinguish bipolar from unipolar depression using deep sequence models, without relying on hand-crafted feature engineering. Through a systematic four-approach investigation, we find a clear answer: during depressive episodes, the motor activity patterns of bipolar and unipolar patients in the Depresjon dataset are statistically indistinguishable. Our most rigorous classifier achieves a best honest accuracy of 63.4% using 48-hour CNN-LSTM windows, barely above the 65.2% majority-class baseline, and direct hypothesis testing confirms negligible group differences ($p = 0.776$, Cohen's $d = -0.060$ [29]). This is a rigorous negative result. It does not indicate that actigraphy is uninformative about mental health more broadly; it specifically rules out wrist activity variability during depressive episodes as a standalone discriminator between bipolar and unipolar depression. The failure of a 223,682-parameter deep network [30] to outperform a 5-feature logistic regression [24] is itself an instructive finding: in clinical machine learning, more complex models do not compensate for insufficient or non-discriminative data. The negative result redirects future research toward concrete and well-motivated alternatives.

Three directions stand out as most promising. First, the Depresjon dataset captures only depressive episodes, but the defining clinical distinction of bipolar disorder is the existence of manic or hypomanic phases [17]. Longitudinal monitoring across full mood episode cycles, including periods of elevated mood, reduced sleep need, and increased goal-directed activity, would provide the class-discriminating signal that depressive-phase recordings cannot. Wehr et al. [4] document that bipolar patients exhibit phase-advanced circadian rhythms most visibly during manic states, not depressive ones, which is precisely the signal that is absent from this dataset. Second, wrist actigraphy alone is a narrow window into the physiology underlying mood disorders. Integrating additional biosignal modalities would substantially enrich the available information: polysomnographic sleep stage data can capture REM-sleep abnormalities that differ between bipolar and unipolar depression at a mechanistic level; heart rate variability reflects autonomic nervous system dysregulation that tracks mood state; and ecological momentary assessment, which captures self-reported mood several times daily, provides a subjective signal that can be aligned with objective sensor data to identify the lag between physiological change and subjective experience. Third, and most fundamentally, any future study targeting the bipolar-versus-unipolar

discrimination task with statistical credibility requires a substantially larger bipolar cohort. To achieve 80% power for detecting a small effect size ($d = 0.2$) at the $\alpha = 0.05$ significance level, approximately 200 bipolar participants are needed [29]. Such a cohort must also reflect demographic diversity in age, sex, medication history, and occupation to ensure that learned features reflect genuine diagnostic signal rather than confounded population differences.

The ethical dimensions of this line of research deserve careful consideration alongside the technical findings. Diagnostic algorithms trained on small, demographically constrained datasets carry the risk of encoding spurious patterns that fail to generalize to broader patient populations, potentially automating misdiagnosis at scale. The Depresjon dataset, with only 8 bipolar participants, is not large enough to characterize the true distribution of bipolar motor activity patterns across the condition's full heterogeneity, and any deployed system trained on it could exhibit systematic failures for subgroups not adequately represented in the sample. Beyond generalization, continuous passive wrist monitoring raises meaningful privacy concerns: the data collected is detailed enough to infer not just mood state but daily routines, social patterns, and behavioral habits, and patients in vulnerable populations may not fully appreciate the scope of what is recorded. Algorithmic accountability is a further concern: clinicians may implicitly defer to model outputs rather than exercising independent judgment, making interpretability and calibration of such systems safety-critical. Our best models reported 88% accuracy while detecting zero bipolar cases, a failure mode that would not be obvious from the headline metric alone. Any future deployment of actigraphy-based classifiers must be accompanied by transparent reporting of sensitivity, specificity, calibration, and explicit uncertainty quantification alongside headline accuracy figures. The path from a promising result on a 55-person dataset to a clinically validated tool is long and uncertain, and intellectual honesty about negative findings such as ours is an essential part of navigating it responsibly.

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